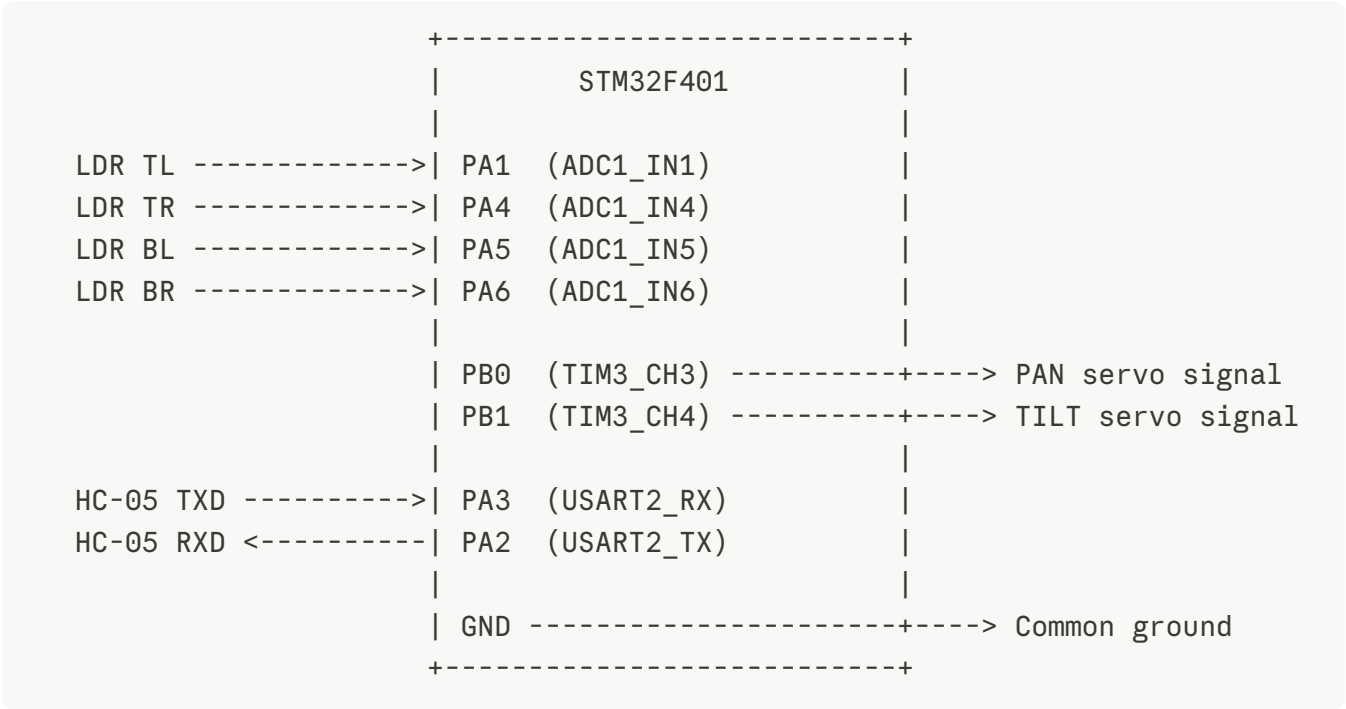
STM32F401 Solar Tracker – Pin Diagram

Pin assignments and wiring reference derived from the firmware: a dual-axis solar tracker with four LDRs, two servos and HC-05 Bluetooth control.

1. Pin Summary

Function	STM32 Pin	Peripheral / Channel	Alt. Function	Connects To
Pan servo (azimuth)	PB0	TIM3_CH3	AF2	Pan servo signal wire
Tilt servo (elevation)	PB1	TIM3_CH4	AF2	Tilt servo signal wire
LDR Top-Left (TL)	PA1	ADC1_IN1	Analog	TL LDR divider output
LDR Top-Right (TR)	PA4	ADC1_IN4	Analog	TR LDR divider output
LDR Bottom-Left (BL)	PA5	ADC1_IN5	Analog	BL LDR divider output
LDR Bottom-Right (BR)	PA6	ADC1_IN6	Analog	BR LDR divider output
Bluetooth TX (to MCU)	PA3	USART2_RX	AF7	HC-05 TXD
Bluetooth RX (from MCU)	PA2	USART2_TX	AF7	HC-05 RXD

2. Block Diagram



3. LDR Layout (Sensor Quadrants)



- Azimuth error = Left average − Right average, which drives the pan servo (PB0).
- Elevation error = Top average − Bottom average, which drives the tilt servo (PB1).

4. Wiring Details

Servos

Servo	Signal	Power	Ground
Pan	PB0	External 5 V supply	Common GND
Tilt	PB1	External 5 V supply	Common GND

- PWM: 50 Hz (20 ms period), pulse 1000–2000 µs.
- Pan range: 0–180°. Tilt range: 0–90°.
- Power the servos from a separate supply, not the MCU pin, and tie its ground to the STM32 ground.

LDR Sensors

Each LDR is wired as a voltage divider; the midpoint goes to the ADC pin. The divider resistor value is not defined in the code (a 10 kΩ fixed resistor is a typical choice).

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3.3V ---[ LDR ]---+---[ R fixed ]--- GND
                  |
                  +--> ADC pin (PA1 / PA4 / PA5 / PA6)
```

- ADC: 12-bit (0–4095), sample time 84 cycles.
- Night threshold: 200. Deadband: 200.

HC-05 Bluetooth Module

HC-05 Pin	STM32 / Supply
VCC	5 V (or 3.3 V, depending on module board)
GND	GND
TXD	PA3 (USART2_RX)
RXD	PA2 (USART2_TX)

- Baud rate: 9600, 8N1.
- Commands: S/s stops automatic tracking; R/r resumes it.
- TX/RX are crossed: module TXD goes to MCU RX, and module RXD goes to MCU TX.
- HC-05 RXD is a 3.3 V logic input. PA2 outputs 3.3 V, so it connects directly.

5. Peripheral Configuration

Peripheral	Setting
System clock	16 MHz (HSI)
TIM3	PSC = 15 (1 MHz tick), ARR = 19999 (50 Hz)
ADC1	Software-triggered, single conversion, 12-bit

Peripheral	Setting
USART2	BRR = 1667, RX interrupt enabled, NVIC priority 2

6. Pin Usage Overview

Port	Pins Used
GPIOA	PA1, PA2, PA3, PA4, PA5, PA6
GPIOB	PB0, PB1

All pins listed are free of conflicts: no two functions share a pin.